



A generalized nonconvex algorithm framework for low-rank and sparse matrix decomposition

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Abstract

The low-rank and sparse matrix decomposition problem is a hot and challenging problem in computer science. In this paper, we consider it as a nonconvex relaxation optimization problem by using a family of nonconvex functions to approximate the rank function and the ℓ_0 -norm in low-rank and sparse matrix decomposition problem, namely, generalized low-rank and sparse matrix decomposition problem. The essence of this paper is to develop an adaptive algorithm framework with parameters updating for the nonconvex relaxation problem. Firstly, we prove the equivalence between the generalized low-rank and sparse matrix decomposition problem and the regularization generalized low-rank and sparse matrix decomposition problem. This means that the optimal solution of generalized low-rank and sparse matrix decomposition problem can be exactly obtained by solving its regularization minimization problem. Secondly, we present a tractable nonconvex algorithm framework to solve the regularization generalized low-rank and sparse matrix decomposition problem. The convergence analysis of the algorithm framework is provided. More importantly, we also define a very powerful parameter-setting strategy to adapt the optimal parameters in iteration of the proposed algorithm framework. Finally, we test the proposed algorithms on some random low-rank and sparse matrix decomposition problems, and the numerical results verified the effectiveness of the proposed algorithms. In addition, we also extend the proposed algorithms to the image denoising and background modeling from surveillance video.

Keywords Low-rank and sparse matrix decomposition · Nonconvex function · regularization Reneralized low-rank and sparse matrix decomposition problem · Nonconvex algorithm framework

1 Introduction

The goal of low-rank and sparse matrix decomposition problem is to decompose a given observed data matrix $D \in \mathbb{R}^{m \times n}$ into a low-rank matrix $L_0 \in \mathbb{R}^{m \times n}$ and a sparse matrix $S_0 \in \mathbb{R}^{m \times n}$, i.e.,

$$D = L_0 + S_0. \quad (1)$$

In mathematics, the low-rank and sparse matrix decomposition problem can be modeled into the following minimization problem:

$$\begin{aligned} \min_{L, S \in \mathbb{R}^{m \times n}} \quad & \text{rank}(L) + \lambda \|S\|_0 \\ \text{s.t.} \quad & D = L + S, \end{aligned} \quad (2)$$

where $\text{rank}(L)$ denotes the rank of matrix L , $\|S\|_0$ denotes the ℓ_0 -norm of matrix S , which counts the number of the nonzero elements of S , and $\lambda > 0$ represents the weighting

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parameter. Without loss of generality, we assume $m \geq n$ in this paper. The low-rank and sparse matrix decomposition problem has been applied in many fields such as image processing [1, 2], video surveillance [3, 4], face recognition [5], Latent semantic indexing [5, 6], and so on.

Unfortunately, the optimization problem (2) is actually NP-hard because of the discrete nature of the ℓ_0 -norm and the discontinuous nature of the rank function. To address this problem, the researchers usually use the following tractable optimization problem to relax the NP-hard problem (2), i.e.,

$$\min_{L, S \in \mathbb{R}^{m \times n}} \|L\|_* + \lambda \|S\|_{\ell_1} \quad (3)$$

$$\text{s.t. } D = L + S,$$

where $\|L\|_* = \sum_{i=1}^n \sigma_i(L)$ denotes the nuclear norm of matrix L , and $\|S\|_{\ell_1} = \sum_{i=1}^m \sum_{j=1}^n |S_{i,j}|$ denotes the ℓ_1 -norm of matrix S . As a convex optimization problem, problem (3) can be efficiently solved by the alternating direction methods [5, 7, 8]. In particular, Candès et al [5] have proved that under some suitable assumptions, problem (3) can exactly recover the low-rank and sparse matrices with high probability.

Although nuclear norm and ℓ_1 -norm simplify the optimization problem (3), they are known to result in biased estimates [9, 10] and perform not very well in decomposing the low-rank and sparse matrices. In fact, the nuclear norm and ℓ_1 -norm are only the loose approximation of rank function and ℓ_0 -norm, but not the tight approximation. With the recent development of nonconvex relaxation approach in matrix rank minimization problems and sparse signal recovery problems, many researchers have shown that the nonconvex functions can serve as better approximation of rank function and ℓ_0 -norm. Indeed, the numerical results have demonstrated that some nonconvex functions outperform the nuclear norm and ℓ_0 -norm in recovering the low-rank matrices and sparse signals (see [11–16]). To fully take the advantages of nonconvex functions, many nonconvex

functions can be used to approximate the rank function and the ℓ_0 -norm in low-rank and sparse matrix decomposition problem by acting on the singular values of matrix L and the elements of matrix S , respectively. This leads to the following generalized low-rank and sparse matrix decomposition problem:

$$\min_{L, S \in \mathbb{R}^{m \times n}} F(\sigma(L)) + \lambda G(S) \quad (4)$$

$$\text{s.t. } D = L + S,$$

where $F(\sigma(L))$ and $G(S)$ are respectively defined as

$$F(\sigma(L)) = \sum_{i=1}^m f_1(\sigma_i(L)), \quad (5)$$

$$G(S) = \sum_{j=1}^n \sum_{i=1}^m f_2(|S_{i,j}|), \quad (6)$$

and $f_i (i = 1, 2)$ represents the suitably chosen non-convex function, σ_i represents the i -th largest singular value of matrix L . Here, $F(\sigma(L))$ is the non-convex function defined on the singular value vector $\sigma(L) : \sigma_1(L) \geq \sigma_2(L) \geq \dots \geq \sigma_n(L) \geq 0$ of matrix L , which approximates the rank function $\text{rank}(L)$, and $G(S)$ is the nonconvex function defined on the element $S_{i,j}$ of matrix S , which approximates the ℓ_0 -norm $\|S\|_0$. In this paper, the nonconvex function $f_i (i = 1, 2)$ satisfies the following assumptions:

1. $f_i : \mathbb{R} \rightarrow \mathbb{R}^+$ is even and continuous;
2. f_i is concave and strictly monotonically increasing on $[0, +\infty)$;
3. $f_i(0) = 0$;
4. $f_i(x)$ is differentiable for all $x \in \mathbb{R}$ except $x = 0$.

Some popular nonconvex functions f_i that satisfy above assumptions are listed in Table 1.

Table 1 Some nonconvex functions f_i

Name	Function $f_i(x)$	Parameter
ℓ_p [14, 17]	$ x ^p$	$0 < p < 1$
Exp [18]	$1 - e^{-\frac{ x }{\delta}}$	$\delta > 0$
Log [19]	$\frac{1}{a} \log(1 + a x)$	$a > 0$
TL1 [15]	$\frac{(a+1) x }{a+ x }$	$a > 0$
Fraction [16]	$\frac{a x }{1+a x }$	$a > 0$
Composite [20]	$1 - e^{-\frac{ x }{\delta(x +\delta)}}$	$\delta > 0$
Arctangent [19]	$\frac{2}{a\sqrt{3}} \left(\tan^{-1} \left(\frac{1+2a x }{\sqrt{3}} \right) - \frac{\pi}{6} \right)$	$a > 0$

Note that the generalized low-rank and sparse matrix decomposition problem (4) is a nonconvex constrained optimization problem, and solving such a problem is highly impractical. Some researchers focus on the following regularization generalized low-rank and sparse matrix decomposition problem [21, 22]:

$$\min_{L, S \in \mathbb{R}^{m \times n}} \left\{ F(\sigma(L)) + \lambda G(S) + \frac{\mu}{2} \|D - L - S\|_F^2 \right\}, \quad (7)$$

where $\mu > 0$ is the regularization parameter. In [21], a proximal average optimization method based on the soft thresholding operator [9] has been applied to solve the regularization generalized low-rank and sparse matrix decomposition problem (7). In [22], a 4-block alternative direction method of multipliers (ADMM) algorithm has been generated to solve the regularization generalized low-rank and sparse matrix decomposition problem (7). It should be emphasized that these two works have a common drawback that they do not provide the optimal selection strategy for the parameters in the model, but only determine the optimal parameters by trial and error method.

Different from the previous work, in this paper we will study a new and comprehensive theoretical and algorithmic framework for the regularization generalized low-rank and sparse matrix decomposition problem (7). The main contributions of our work are summarized as follows:

- We establish the equivalence between the generalized low-rank and sparse matrix decomposition problem (4) and its regularization problem, i.e., regularization generalized low-rank and sparse matrix decomposition problem (7), which means that the generalized low-rank and sparse matrix decomposition problem (4) can be exactly addressed by solving the regularization generalized low-rank and sparse matrix decomposition problem (7).
- We propose a tractable algorithm framework for solving the regularization generalized low-rank and sparse matrix decomposition (7). In particular, a convergence analysis for the proposed algorithm is given. More importantly, we provide a very powerful parameter-setting strategy for the optimal parameters μ and λ .

The rest of this paper is organized as follows. In Section 2, we establish the equivalence between the generalized low-rank and sparse matrix decomposition problem (4) and the regularization generalized low-rank and sparse matrix decomposition problem (7). In Section 3, we propose a tractable algorithm framework to solve the regularization generalized low-rank and sparse matrix decomposition problem (7). The convergence analysis of the proposed

algorithm framework is also proved. In Section 4, a very parameter-setting strategy for the optimal parameters μ and λ is defined. The numerical results are presented in Section 5. Finally, we conclude this paper in Section 6.

2 Equivalence analysis

In this section, we show that the generalized low-rank and sparse matrix decomposition problem (4) can be exactly solved by solving the regularization generalized low-rank and sparse matrix decomposition problem (7).

Theorem 1 For any $\mu > 0$, the set of the optimal solutions of regularization generalized low-rank and sparse matrix decomposition problem (7) is nonempty. Let $\{\mu_k\}$ be an increasing sequence of positive numbers with $\mu_k \rightarrow +\infty$, and (L^{μ_k}, S^{μ_k}) be an optimal solution of regularization generalized low-rank and sparse matrix decomposition problem (7) with $\mu = \mu_k$. Suppose generalized low-rank and sparse matrix decomposition problem (4) is feasible, then any accumulation point of (L^{μ_k}, S^{μ_k}) is an optimal solution of generalized low-rank and sparse matrix decomposition problem (4).

Proof Since $V_{\lambda, \mu}(L, S) := F(\sigma(L)) + \lambda G(S) + \frac{\mu}{2} \|D - L - S\|_F^2 \geq 0$ and $V_{\lambda, \mu}(L, S) \leq F(\sigma(L)) + \lambda G(S) + \frac{\mu}{2} \|D\|_F^2 + \|L\|_F^2 + \|S\|_F^2 \rightarrow +\infty$ if $\|L\|_F \rightarrow +\infty$ and $\|S\|_F \rightarrow +\infty$, the set of the optimal solutions of regularization generalized low-rank and sparse matrix decomposition problem (7) is nonempty.

Therefore, the set of the optimal solutions of problem (7) with $\mu = \mu_k$ is also nonempty. This means that the optimal solution (L^{μ_k}, S^{μ_k}) exists. Suppose that problem (4) is feasible and $(\hat{L}, \hat{S})$ is any feasible point, then $D = \hat{L} + \hat{S}$. Since (L^{μ_k}, S^{μ_k}) is the optimal solution of problem (7) with $\mu = \mu_k$, it follows that

$$\begin{aligned} F(\sigma(L^{\mu_k})) + \lambda G(S^{\mu_k}) + \frac{\mu_k}{2} \|D - L^{\mu_k} - S^{\mu_k}\|_F^2 \\ \leq F(\sigma(\hat{L})) + \lambda G(\hat{S}) + \frac{\mu_k}{2} \|D - \hat{L} - \hat{S}\|_F^2 \\ = F(\sigma(\hat{L})) + \lambda G(\hat{S}). \end{aligned}$$

Hence,

$$F(\sigma(L^{\mu_k})) + \lambda G(S^{\mu_k}) \leq F(\sigma(\hat{L})) + \lambda G(\hat{S}),$$

which shows that $\{(L^{\mu_k}, S^{\mu_k})\}$ is bounded, uniformly in μ . Thus, the set $\{(L^{\mu_k}, S^{\mu_k})\}$ has at least one accumulation point. Let $(L^\sharp, S^\sharp)$ be any accumulation point of $\{(L^{\mu_k}, S^{\mu_k})\}$. From

$$\begin{aligned} &\leq \frac{2}{\mu_k} \left(F(\sigma(L^{\mu_k})) + \lambda G(S^{\mu_k}) + \frac{\mu_k}{2} \|D - L^{\mu_k} - S^{\mu_k}\|_F^2 \right) \\ &\leq \frac{2}{\mu_k} \left(F(\sigma(\hat{L})) + \lambda G(\hat{S}) \right) \end{aligned}$$

and $\mu_k \rightarrow +\infty$, we have $D = L^\sharp + S^\sharp$, that is, $(L^\sharp, S^\sharp)$ is a feasible point of problem (4). By

$$F(\sigma(L^{\mu_k})) + \lambda G(S^{\mu_k}) \leq F(\sigma(\hat{L})) + \lambda G(\hat{S}),$$

we have

$$F(\sigma(L^\sharp)) + \lambda G(S^\sharp) \leq F(\sigma(\hat{L})) + \lambda G(\hat{S}).$$

Furthermore, by the arbitrariness of $(\hat{L}, \hat{S})$, we can get that $(L^\sharp, S^\sharp)$ is an optimal solution of problem (4). $\square$

3 Tractable nonconvex algorithm framework

In this section, a tractable nonconvex algorithm framework is designed to solve the problem (7). Moreover, a powerful parameter-setting strategy is also defined to set the optimal parameters for the proposed algorithm framework.

3.1 Proximal operator

To begin with, we first introduce the proximal operator of f_i , which plays a key role in solving the problem (7).

Obviously, the nonconvex function f_i defined in this paper is proper and lower semicontinuity. According to [23, 1.22 Definition], we can give the definition of the proximal operator of nonconvex function f_i .

Definition 1 For nonconvex function f_i , its proximal operator denoted $p_{f_i, \eta} : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$p_{f_i, \eta}(t) = \arg \min_{x \in \mathbb{R}} \left\{ \frac{1}{2}(x - t)^2 + \eta f_i(x) \right\}, \quad (8)$$

where $\eta > 0$.

For such a proximal operator, we summarize some properties of $p_{f_i, \eta}$ below.

Property 1 The proximity operator $p_{f_i, \lambda}$ defined in (8) is odd, i.e., $p_{f_i, \lambda}(-t) = -p_{f_i, \lambda}(t)$.

Proof Build the coordinate system with horizontal x -axis and the vertical y -axis. Let $y_1 = \frac{1}{2}(x - t)^2 + \lambda f_i(x)$ and $y_2 = \frac{1}{2}(x + t)^2 + \lambda f_i(x)$. Obviously, functions y_1 and y_2

are symmetric about the y -axis. Therefore, the minimizers of y_1 and y_2 are also symmetric about the y -axis. The proof is complete. $\square$

Property 2 The proximal operator $p_{f_i, \eta}$ defined in (8) is increasing on $\mathbb{R}$.

Proof Let $\gamma_1 \in \mathbb{R}$ and $\gamma_2 \in \mathbb{R}$ be two different numbers. Suppose that

$$p_{f_i, \eta}(\gamma_1) = \arg \min_{x \in \mathbb{R}} \left\{ \frac{1}{2}(x - \gamma_1)^2 + \eta f_i(x) \right\},$$

$$p_{f_i, \eta}(\gamma_2) = \arg \min_{x \in \mathbb{R}} \left\{ \frac{1}{2}(x - \gamma_2)^2 + \eta f_i(x) \right\}.$$

Then, we have

$$\begin{aligned} &\frac{1}{2}(p_{f_i, \eta}(\gamma_2) - \gamma_1)^2 + \eta f_i(p_{f_i, \eta}(\gamma_2)) \geq \\ &\frac{1}{2}(p_{f_i, \eta}(\gamma_1) - \gamma_1)^2 + \eta f_i(p_{f_i, \eta}(\gamma_1)) \end{aligned} \quad (9)$$

and

$$\begin{aligned} &\frac{1}{2}(p_{f_i, \eta}(\gamma_1) - \gamma_2)^2 + \eta f_i(p_{f_i, \eta}(\gamma_1)) \geq \\ &\frac{1}{2}(p_{f_i, \eta}(\gamma_2) - \gamma_2)^2 + \eta f_i(p_{f_i, \eta}(\gamma_2)). \end{aligned} \quad (10)$$

Combining (9) and (10), we obtain

$$(p_{f_i, \eta}(\gamma_2) - \gamma_1)^2 + (p_{f_i, \eta}(\gamma_1) - \gamma_2)^2 \geq (p_{f_i, \eta}(\gamma_1) - \gamma_1)^2 + (p_{f_i, \eta}(\gamma_2) - \gamma_2)^2.$$

After simplification, we obtain

$$(p_{f_i, \eta}(\gamma_1) - p_{f_i, \eta}(\gamma_2))(\gamma_1 - \gamma_2) \geq 0.$$

This completes the proof. $\square$

In addition, we show that under milder assumptions, the operator $p_{f_i, \eta}$ can be expressed as a thresholding function. To formulate the operator $p_{f_i, \eta}$, we define

$$P_{f_i}^\eta = \left\{ t \mid \frac{1}{2}t^2 = \min_{x > 0} \left\{ \frac{1}{2}(x - t)^2 + \eta f_i(x) \right\}, t > 0 \right\}. \quad (11)$$

For any $t^* \in P_{f_i}^\eta$, it is obvious that t^* is determined by nonconvex function f_i and parameter η . Moreover, it is easy to verify that t^* is increasing with respect to η . Therefore, we can write t^* as $t_{f_i}(\eta)$. The following theorem presents the expression of operator $p_{f_i, \eta}$.

Theorem 2 If there exists $t_{f_i}(\eta) \in P_{f_i}^\eta$ such that the following conditions hold:

1. $|t| \leq t_{f_i}(\eta) \iff \frac{1}{2}t^2 \leq \min_{x \neq 0} \left\{ \frac{1}{2}(x-t)^2 + \eta f_i(x) \right\};$
2. $|t| > t_{f_i}(\eta) \iff \frac{1}{2}t^2 > \min_{x \neq 0} \left\{ \frac{1}{2}(x-t)^2 + \eta f_i(x) \right\}.$

Then, the proximal operator $p_{f_i,\eta}$ can be expressed as:

$$p_{f_i,\eta}(t) = \begin{cases} s_{f_i,\eta}(|t|) \operatorname{sign}(t), & |t| > t_{f_i}(\eta) \\ 0, & |t| \leq t_{f_i}(\eta) \end{cases} \quad (12)$$

where $s_{f_i,\eta}(t)$ is one real root of equation $x - t + \eta f'_i(x) = 0$ with $x > 0$, and $t_{f_i}(\eta)$ represents the threshold value.

Proof Let $F(x) = \frac{1}{2}(x-t)^2 + \eta f_i(x)$. To derive the minimum of $F(x)$, we distinguish two cases, $x \neq 0$ and $x = 0$. Therefore, $F(x)$ can be rewritten as

$$F(x) = \begin{cases} \frac{1}{2}(x-t)^2 + \eta f_i(x), & x \neq 0; \\ \frac{1}{2}t^2, & x = 0. \end{cases} \quad (13)$$

Without loss of generality, we set $x > 0$ (in $x \neq 0$ case) and $t > 0$, then (13) can be further rewritten as

$$F(x) = \begin{cases} \frac{1}{2}(x-t)^2 + \eta f_i(x), & x > 0; \\ \frac{1}{2}t^2, & x = 0. \end{cases} \quad (14)$$

Based on assumptions, we can immediately obtain the following conclusions:

- When $t \leq t_{f_i}(\eta)$, the minimum of $F(x)$ is attained at $x = 0$.
- When $t > t_{f_i}(\eta)$, the minimum of $F(x)$ is attained at $x = s_{f_i,\eta}(t)$, where $s_{f_i,\eta}(t)$ is one real root of $x - t + \eta f'_i(x) = 0$.

By Property 1, we can immediately get the formulation (12). The proof is thus completed. $\square$

For the most functions f_i , the equation $x - t + \eta f'_i(x) = 0$ is a transcendental equation and has no explicit solution. This indicates that the proximal operator $p_{f_i,\eta}$ defined in (8) has no explicit expression for the most functions f_i . In fact, only a very few nonconvex functions f_i can guarantee their proximal operators have the explicit expressions. In the following, we list some special nonconvex functions f_i with the explicit expressions of proximal operators.

- For $f_i(x) = |x|^{\frac{1}{2}}$ [13], the proximal operator $p_{f_i,\eta}$ defined in (8) can be analytically expressed as

$$p_{f_i,\eta}(t) = \begin{cases} s_{f_i,\eta}(|t|) \operatorname{sign}(t), & |t| > \frac{3}{2}\eta^{\frac{2}{3}} \\ 0, & |t| \leq \frac{3}{2}\eta^{\frac{2}{3}} \end{cases} \quad (15)$$

where $s_{f_i,\eta}(|t|)$ is defined by

$$s_{f_i,\eta}(|t|) = \frac{2}{3}|t| \left(1 + \cos \left(\frac{2\pi}{3} - \frac{2}{3}\varphi_\eta(t) \right) \right)$$

with $\varphi_\eta(t) = \arccos \left(\frac{\eta}{4} \left(\frac{|t|}{3} \right)^{-\frac{3}{2}} \right)$.

- For $f_i(x) = |x|^{\frac{2}{3}}$ [14], the proximal operator $p_{f_i,\eta}$ defined in (8) can be analytically expressed as

$$p_{f_i,\eta}(t) = \begin{cases} s_{f_i,\eta}(|t|) \operatorname{sign}(x), & |t| > \frac{2\sqrt[3]{24}}{3}\eta^{\frac{3}{4}} \\ 0, & |t| \leq \frac{2\sqrt[3]{24}}{3}\eta^{\frac{3}{4}} \end{cases} \quad (16)$$

where $s_{f_i,\eta}(|t|)$ is defined by

$$s_{f_i,\eta}(|t|) = \left(\frac{|\Phi_\eta(t)| + \sqrt{\frac{2|t|}{|\Phi_\eta(t)|} - |\Phi_\eta(t)|^2}}{2} \right)^3$$

with $\Phi_\eta(t) = \frac{2}{\sqrt{3}}\eta^{\frac{1}{4}} \left(\cosh \left(\frac{1}{3} \operatorname{arccosh} \left(\frac{27}{16}\eta^{-\frac{3}{2}}t^2 \right) \right) \right)^{\frac{1}{2}}$.

- For $f_i(x) = \frac{1}{a} \log(1 + a|x|)$ [19], the proximity operator $p_{f_i,\eta}$ defined in (8) can be analytically expressed as

$$p_{f_i,\eta}(t) = \begin{cases} s_{f_i,\eta}(|t|) \operatorname{sign}(x), & |t| > \eta \\ 0, & |t| \leq \eta \end{cases} \quad (17)$$

where $s_{f_i,\eta}(|t|)$ is defined by

$$s_{f_i,\eta}(|t|) = \left(\frac{|y|}{2} - \frac{1}{2a} + \sqrt{\left(\frac{|y|}{2} + \frac{1}{2a} \right)^2 - \frac{\eta}{a}} \right).$$

- For $f_i(x) = \frac{(a+1)|x|}{a+|x|}$ [15], the proximal operator $p_{f_i,\eta}$ defined in (8) can be analytically expressed as

$$p_{f_i,\eta}(t) = \begin{cases} s_{f_i,\eta}(|t|) \operatorname{sign}(t), & |t| > t_{f_i}(\eta) \\ 0, & |t| \leq t_{f_i}(\eta) \end{cases} \quad (18)$$

where $s_{f_i,\eta}(|t|)$ is defined by

$$s_{f_i,\eta}(|t|) = \left(\frac{2}{3}(a + |t|) \cos \left(\frac{\varphi(t)}{3} \right) - \frac{2a}{3} + \frac{|x|}{3} \right)$$

with $\varphi(t) = \arccos \left(1 - \frac{27\eta a(a+1)}{2(a+|t|)^3} \right)$, and the parameter $t_{f_i}(\eta)$ satisfies

$$t_{f_i}(\eta) = \begin{cases} \frac{\eta(a+1)}{a}, & \eta \leq \frac{a^2}{2(a+1)} \\ \sqrt{2\eta(a+1)} - \frac{a}{2}, & \eta > \frac{a^2}{2(a+1)}. \end{cases}$$

- For $f_i(x) = \frac{a|x|}{1+a|x|}$ [16], the proximal operator $p_{f_i,\eta}$ defined in (8) can be analytically expressed as

$$p_{f_i,\eta}(t) = \begin{cases} s_{f_i,\eta}(|t|)\text{sign}(t), & |t| > t_{f_i}(\eta) \\ 0, & |t| \leq t_{f_i}(\eta) \end{cases} \quad (19)$$

where $s_{f_i,\eta}(|t|)$ is defined by

$$s_{f_i,\eta}(|t|) = \left(\frac{1+a|t|}{3a} \left(1 + 2 \cos \left(\frac{\phi(t)}{3} - \frac{\pi}{3} \right) \right) - \frac{1}{a} \right)$$

with $\phi(t) = \arccos \left(\frac{27\eta a^2}{2(1+a|t|)^3} - 1 \right)$, and the parameter $t_{f_i}(\eta)$ satisfies

$$t_{f_i}(\eta) = \begin{cases} \eta a, & \eta \leq \frac{1}{2\eta^2} \\ \sqrt{2\eta} - \frac{1}{2a}, & \eta > \frac{1}{2a^2} \end{cases}$$

3.2 Matrix proximal operators

Based on the proximal operator $p_{f_i,\eta}$ defined in (8), matrix rank proximal operator and matrix element proximal operator can be defined as follows.

Definition 2 (*Matrix rank proximal operator*) For any $\eta > 0$ and matrix $Y \in \mathbb{R}^{m \times n}$, let

$$Y = U \begin{bmatrix} \text{Diag}(\sigma(Y)) \\ 0_{(m-n) \times n} \end{bmatrix} V^\top$$

be the singular value decomposition (SVD) of matrix $Y \in \mathbb{R}^{m \times n}$, where $0_{(m-n) \times n} \in \mathbb{R}^{(m-n) \times n}$ is a zero matrix and $\text{Diag}(\sigma(Y)) \in \mathbb{R}^{n \times n}$ is the diagonal matrix of the singular value vector $\sigma(Y)$ of matrix Y arranged in nonincreasing order $\sigma(Y) : \sigma_1(Y) \geq \sigma_2(Y) \geq \dots \geq \sigma_n(Y) \geq 0$, the matrix rank proximal operator $\mathcal{H}_{f_1,\eta} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ is defined as

$$\mathcal{H}_{f_1,\eta}(Y) = U \begin{bmatrix} \text{Diag}(H_{f_1,\eta}(\sigma(Y))) \\ 0_{(m-n) \times n} \end{bmatrix} V^\top, \quad (20)$$

where

$$H_{f_1,\eta}(\sigma(Y)) = [p_{f_1,\eta}(\sigma_1(Y)), p_{f_1,\eta}(\sigma_2(Y)), \dots, p_{f_1,\eta}(\sigma_n(Y))]^\top.$$

Note that the matrix rank thresholding operator $\mathcal{H}_{f_1,\eta}$ is defined on the singular value vector of a matrix through

operator $p_{f_1,\eta}$. Hence, the rank of $\mathcal{H}_{f_1,\eta}(Y)$ is determined by the characteristics of operator $p_{f_1,\eta}$. Indeed, if some singular value is lower than the threshold value $t_{f_1}(\eta)$, the operator $p_{f_1,\eta}$ can shrink the singular value to zero. As a result, the matrix rank thresholding operator $\mathcal{H}_{f_1,\eta}$ can generate a low-rank matrix.

Definition 3 (*Matrix element proximal operator*) For any $\eta > 0$ and matrix $Y \in \mathbb{R}^{m \times n}$, the matrix element proximal operator $\mathcal{G}_{f_2,\eta} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ is defined as

$$\mathcal{G}_{f_2,\eta}(Y) = \begin{bmatrix} p_{f_2,\eta}(Y_{1,1}) & \cdots & p_{f_2,\eta}(Y_{1,n}) \\ \vdots & \ddots & \vdots \\ p_{f_2,\eta}(Y_{m,1}) & \cdots & p_{f_2,\eta}(Y_{m,n}) \end{bmatrix}. \quad (21)$$

Note that $\mathcal{G}_{f_2,\eta}$ is defined on the elements of a matrix through the operator $p_{f_2,\eta}$. Thus the sparsity of matrix $\mathcal{G}_{f_2,\eta}(Y)$ is determined by the characteristics of operator $p_{f_2,\eta}$. That is, if the elements of matrix Y are lower than the threshold value $t_{f_2}(\eta)$ in magnitude, the operator $p_{f_2,\eta}$ can shrink these elements to zero. Therefore, the matrix element proximal operator $\mathcal{G}_{f_2,\eta}$ can generate a sparse matrix.

In order to use operators $\mathcal{H}_{f_1,\eta}$ and $\mathcal{G}_{f_2,\eta}$ to solve the problem (7), we need prove the following results.

Theorem 3 Suppose $Y \in \mathbb{R}^{m \times n}$ admits the SVD as

$$Y = U \begin{bmatrix} \text{Diag}(\sigma(Y)) \\ 0_{(m-n) \times n} \end{bmatrix} V^\top$$

and

$$\mathcal{H}_{f_1,\eta}(Y) = U \begin{bmatrix} \text{Diag}(H_{f_1,\eta}(\sigma(Y))) \\ 0_{(m-n) \times n} \end{bmatrix} V^\top,$$

then

$$\mathcal{H}_{f_1,\eta}(Y) = \arg \min_{L \in \mathbb{R}^{m \times n}} \left\{ \frac{1}{2} \|L - Y\|_F^2 + \eta F(\sigma(L)) \right\}. \quad (22)$$

Proof The proof is similar to the proof of [12], Theorem 3.1], so it is omitted here. $\square$

Theorem 4 For a given $Y \in \mathbb{R}^{m \times n}$ and matrix function $G(S)$ defined in (6), the following equality holds.

$$\mathcal{G}_{f_2,\eta}(Y) = \arg \min_{S \in \mathbb{R}^{m \times n}} \left\{ \frac{1}{2} \|S - Y\|_F^2 + \eta G(S) \right\}. \quad (23)$$

Proof Obviously, the problem

$$\min_{X \in \mathbb{R}^{m \times n}} \left\{ \frac{1}{2} \|S - Y\|_F^2 + \eta G(S) \right\} \quad (24)$$

is equivalent to

$$\min_{X \in \mathbb{R}^{m \times n}} \left\{ \sum_{j=1}^n \sum_{i=1}^m \left(\frac{1}{2} |S_{i,j} - Y_{i,j}|^2 + \eta f_2(|S_{i,j}|) \right) \right\}. \quad (25)$$

Note that the objective function in problem (25) is separable, hence, solving problem (25) is equivalent to solving the following mn subproblems, for $i \in \{1, 2, \dots, m\}$, $j \in \{1, 2, \dots, n\}$,

$$\min_{S_{i,j} \in \mathbb{R}} \left\{ \frac{1}{2} |S_{i,j} - Y_{i,j}|^2 + \eta f_2(|S_{i,j}|) \right\}. \quad (26)$$

By Definition 1, the optimal solution $S_{i,j}^*$ of problem (26) is given by

$$S_{i,j}^* = p_{f_2, \eta}(Y_{i,j}).$$

Therefore, it follows that $S^* = \mathcal{G}_{f_2, \eta}(Y)$ is the optimal solution to the problem (24). The proof is completed. $\square$

From Theorem 3 and Theorem 4, we can see that the matrix rank proximal operator $\mathcal{H}_{f_1, \eta}$ and the matrix element proximal operator $\mathcal{G}_{f_2, \eta}$ are in fact the proximal operators of $F(\sigma(L))$ and $G(S)$, respectively. They are actually the natural generalization of the operators $p_{f_1, \eta}$ and $p_{f_2, \eta}$ from scalars to matrices.

3.3 A tractable nonconvex algorithm framework

This subsection presents a tractable nonconvex algorithm framework to solve the problem (7) based on Theorem 3 and Theorem 4. The tractable nonconvex algorithm framework adopts the alternating technique and iteratively updates both the low-rank and sparse components in proximal form. Essentially speaking, it is a 2-block ADMM algorithm.

For any $\lambda, \mu \in (0, +\infty)$, let

$$V_{\lambda, \mu}(L, S) = F(\sigma(L)) + \lambda G(S) + \frac{\mu}{2} \|D - L - S\|_F^2, \quad (27)$$

the nonconvex algorithm framework for solving the problem (7) consists of the iterations

$$\begin{cases} L^{k+1} = \arg \min_{L \in \mathbb{R}^{m \times n}} V_{\lambda, \mu}(L, S^k), \\ S^{k+1} = \arg \min_{S \in \mathbb{R}^{m \times n}} V_{\lambda, \mu}(L^{k+1}, S). \end{cases} \quad (28)$$

After simplification, iteration (28) can be rewritten as

$$\begin{cases} L^{k+1} = \arg \min_{L \in \mathbb{R}^{m \times n}} \left\{ \frac{1}{2} \|L - (D - S^k)\|_F^2 + \frac{1}{\mu} F(\sigma(L)) \right\}, \\ S^{k+1} = \arg \min_{S \in \mathbb{R}^{m \times n}} \left\{ \frac{1}{2} \|S - (D - L^{k+1})\|_F^2 + \frac{\lambda}{\mu} G(S) \right\}. \end{cases} \quad (29)$$

By Theorem 3 and Theorem 4, the optimal solutions L^{k+1} and S^{k+1} can be expressed as

$$\begin{cases} L^{k+1} = \mathcal{H}_{f_1, \frac{1}{\mu}}(D - S^k), \\ S^{k+1} = \mathcal{G}_{f_2, \frac{\lambda}{\mu}}(D - L^{k+1}), \end{cases} \quad (30)$$

where $\mathcal{H}_{f_1, \frac{1}{\mu}}(\cdot)$ is obtained by replacing η with $\frac{1}{\mu}$ in $\mathcal{H}_{f_1, \eta}(\cdot)$, and $\mathcal{G}_{f_2, \frac{\lambda}{\mu}}(\cdot)$ is obtained by replacing η with $\frac{\lambda}{\mu}$ in $\mathcal{G}_{f_2, \eta}(\cdot)$.

In this paper, we call iteration (30) the alternating iterative framework (AIF) algorithm and it is described in Algorithm 1.

Algorithm 1 AIF algorithm.

```

1: Input:  $\lambda > 0, \mu > 0$ 
2: Initialize:  $S^0 \in \mathbb{R}^{m \times n}$ 
3: while not converged do
4:    $L^{k+1} = \mathcal{H}_{f_1, \frac{1}{\mu}}(D - S^k)$ 
5:    $S^{k+1} = \mathcal{G}_{f_2, \frac{\lambda}{\mu}}(D - L^{k+1})$ 
6:    $k = k + 1$ 
7: end while
8: Output:  $L^*, S^*$ 

```

Next, let us analyze the convergence of AIF algorithm.

Theorem 5 Let $\{(L^k, S^k)\}$ be the sequence generated by AIF algorithm. Then $\{(L^k, S^k)\}$ is a minimization sequence, and $V_{\lambda, \mu}(L^k, S^k)$ converges to $V_{\lambda, \mu}(L^*, S^*)$, where (L^*, S^*) is any accumulation point of sequence $\{(L^k, S^k)\}$.

Proof According to the definition of $V_{\lambda, \mu}(L, S)$, we have

$$V_{\lambda, \mu}(L^k, S^{k-1}) = F(\sigma(L^k)) + \lambda G(S^{k-1}) + \frac{\mu}{2} \|D - L^k - S^{k-1}\|_F^2,$$

$$V_{\lambda, \mu}(L^k, S^k) = F(\sigma(L^k)) + \lambda G(S^k) + \frac{\mu}{2} \|D - L^k - S^k\|_F^2.$$

By the optimality of S^k , we have

$$V_{\lambda, \mu}(L^k, S^{k-1}) \geq V_{\lambda, \mu}(L^k, S^k).$$

On the other hand,

$$V_{\lambda,\mu}(L^k, S^k) = F(\sigma(L^k)) + \lambda G(S^k) + \frac{\mu}{2} \|D - L^k - S^k\|_F^2,$$

$$V_{\lambda,\mu}(L^{k+1}, S^k) = F(\sigma(L^{k+1})) + \lambda G(S^k) + \frac{\mu}{2} \|D - L^{k+1} - S^k\|_F^2.$$

Similarly, by the optimality of L^{k+1} , we have

$$V_{\lambda,\mu}(L^k, S^k) \geq V_{\lambda,\mu}(L^{k+1}, S^k).$$

Therefore, we have

$$V_{\lambda,\mu}(L^1, S^0) \geq V_{\lambda,\mu}(L^1, S^1) \geq V_{\lambda,\mu}(L^2, S^1) \geq V_{\lambda,\mu}(L^2, S^2) \geq \dots \\ \geq V_{\lambda,\mu}(L^k, S^{k-1}) \geq V_{\lambda,\mu}(L^k, S^k) \geq V_{\lambda,\mu}(L^{k+1}, S^k) \geq \dots$$

That is, $\{(L^k, S^k)\}$ is a minimization sequence of the objective function $V_{\lambda,\mu}(L, S)$, and the sequence $\{V_{\lambda,\mu}(L^k, S^k)\}$ is monotonically decreasing. Since $\{V_{\lambda,\mu}(L^k, S^k)\}$ is bounded from below, the sequence $\{V_{\lambda,\mu}(L^k, S^k)\}$ converges to a constant V^* . Because $\{(L^k, S^k)\} \subset \{(L, S) : V_{\lambda,\mu}(L, S) \leq V_{\lambda,\mu}(L^1, S^0)\}$ which is bounded, it follows that the sequence $\{(L^k, S^k)\}$ is bounded. Therefore, the sequence $\{(L^k, S^k)\}$ has at least one accumulation point. Let (L^*, S^*) be an accumulation point of sequence $\{(L^k, S^k)\}$. By the continuity of $V_{\lambda,\mu}(L, S)$ and the convergence of sequence $\{V_{\lambda,\mu}(L^k, S^k)\}$, we can get that $V_{\lambda,\mu}(L^k, S^k) \rightarrow V^* = V_{\lambda,\mu}(L^*, S^*)$ as $k \rightarrow \infty$. This completes the proof. $\square$

4 Optimal parameters and adaptive algorithm

In this section, we discuss the problem of optimal parameters setting in our algorithm. As we all know, the performance of the algorithm depends seriously on the setting of parameters. However, the setting of the optimal parameters is a very challenging problem due to the lack of a proper optimal rule. Nevertheless, when some prior information can be observed well for the problem, it is possible to set the parameters more reasonably.

4.1 Setting of optimal regularization parameter μ^*

From Theorem 1, we know the optimal solution of problem (7) also solves the problem (4) when the regularization

parameter $\mu \rightarrow +\infty$. This gives us a guidance for setting the optimal regularization parameter μ^* . In this paper, we prefer to use possibly different regularization parameters μ^k for each iteration in AIF algorithm. Therefore, we can update the optimal regularization parameter μ^* in AIF algorithm by

$$\mu_{k+1} = \rho \mu_k, \quad (31)$$

where $\rho > 1$ is a given positive constant.

4.2 Setting of optimal weighting parameter λ^*

According to AIF algorithm, we can see that the setting of weighting parameter λ has a significant impact on the quality of sparse matrix S^{k+1} generated by

$$S^{k+1} = \mathcal{G}_{f_2, \frac{\lambda}{\mu}}(D - L^{k+1}). \quad (32)$$

In the following, we define a very powerful parameter-setting strategy for the setting of the optimal weighting parameter λ^* . To make it clear, we assume (L^*, S^*) is the optimal solution of problem (7) and the AIF algorithm converges to (L^*, S^*) , then

$$S^* = \mathcal{G}_{f_2, \frac{\lambda^*}{\mu^*}}(D - L^*). \quad (33)$$

If the sparsity is known for the optimal sparse solution S^* to the problem (7), the sparse information can be used to develop a very powerful parameter-setting strategy for the optimal weighting parameter λ^* . Specifically, let us suppose that the sparsity of the optimal solution S^* to the problem (7) is r , i.e., $\|S^*\|_0 = r$. Thus, the sparse matrix part required to be solved is restricted to the submanifold $\Gamma_r = \{S \in \mathbb{R}^{m \times n} \mid \|S\|_0 = r\}$. Rearrange the absolute value of elements of matrix $D - L^*$ as the vector $v^* \in \mathbb{R}^{mn}$ in nonincreasing order:

$$v^* : v_1^* \geq v_2^* \geq \dots \geq v_{mn}^* \geq 0.$$

Then, by Theorem 2 and (33), the following inequalities hold:

$$v_j^* > t_{f_2} \left(\frac{\lambda^*}{\mu^*} \right) \Leftrightarrow j \in \{1, 2, \dots, r\}$$

$$v_l^* \leq t_{f_2} \left(\frac{\lambda^*}{\mu^*} \right) \Leftrightarrow l \in \{r+1, r+2, \dots, mn\}$$

which implies that

$$v_{r+1}^* \leq t_{f_2} \left(\frac{\lambda^*}{\mu^*} \right) < v_r^*. \quad (34)$$

If the threshold value function t_{f_2} is invertible and monotonically increasing, we can get that

$$\mu^* t_{f_2}^{-1}(v_{r+1}^*) \leq \lambda^* < \mu^* t_{f_2}^{-1}(v_r^*). \quad (35)$$

The above estimation (35) provides an exact location for the optimal weighting parameter λ^* , and the optimal weighting parameter λ^* can be taken as

$$\lambda^* = \mu^* \left[(1 - \gamma) t_{f_2}^{-1}(v_{r+1}^*) + \gamma t_{f_2}^{-1}(v_r^*) \right] \quad (36)$$

with any $\gamma \in [0, 1]$. Taking $\gamma = 0$, this leads to a most reliable choice of optimal weighting parameter λ^* specified by

$$\lambda^* = \mu^* t_{f_2}^{-1}(v_{r+1}^*). \quad (37)$$

It is worth mentioning that we did not know the optimal solution L^* and the optimal regularization parameter μ^* in practice, so we cannot obtain the optimal weighting parameter λ^* through (37). Actually we can approximate L^* by L^{k+1} in $D - L^*$, and approximate μ^* by μ^k defined in (31). Therefore, in each iteration of AIF algorithm, we can take the optimal weighting parameter λ^* as

$$\lambda_{k+1}^* = \mu_k t_{f_2}^{-1}(v_{r+1}^{k+1}), \quad (38)$$

where v_{r+1}^k represents the $(r + 1)$ -th component of vector v^k which is obtained by rearranging the absolute value of elements of matrix $D - L^{k+1}$ as the vector $v^{k+1} \in \mathbb{R}^{mn}$ with the elements listed in nonincreasing order:

$$v^{k+1} : v_1^{k+1} \geq v_2^{k+1} \geq \dots \geq v_{mn}^{k+1} \geq 0.$$

When doing so, the AIF algorithm will adapt to the choice of the optimal weighting parameter in each iteration.

According to (38), we can see that the expression of λ_{k+1}^* is uniquely determined by the function f_2 , and different f_2 defines the different expression of λ_{k+1}^* .

- For $f_2(x) = |x|^{\frac{1}{2}}$, we have

$$t_{f_2} \left(\frac{\lambda}{\mu} \right) = \frac{3}{2} \left(\frac{\lambda}{\mu} \right)^{\frac{2}{3}}.$$

Then, by (37), the optimal weighting parameter λ^* in each iteration of algorithm can be set as

$$\lambda_{k+1}^* = \frac{\sqrt{24}(v_{r+1}^{k+1})^{\frac{3}{2}} \mu_k}{9}. \quad (39)$$

- For $f_2(x) = |x|^{\frac{2}{3}}$, we have

$$t_{f_2} \left(\frac{\lambda}{\mu} \right) = \frac{2\sqrt[4]{24}}{3} \left(\frac{\lambda}{\mu} \right)^{\frac{3}{4}}.$$

Then, by (37), the optimal weighting parameter λ^* in each iteration of algorithm can be set as

$$\lambda_{k+1}^* = \frac{3(v_{r+1}^{k+1})^{\frac{4}{3}} \mu_k}{4\sqrt[3]{2}}. \quad (40)$$

- For $f_2(x) = \frac{1}{a} \log(1 + a|x|)$ [19], we have

$$t_{f_2} \left(\frac{\lambda}{\mu} \right) = \frac{\lambda}{\mu}.$$

Then, by (37), the optimal weighting parameter λ^* in each iteration of algorithm can be set as

$$\lambda_{k+1}^* = v_{r+1}^{k+1} \mu_k. \quad (41)$$

- For $f_2(x) = \frac{(a+1)|x|}{a+|x|}$ [15], we have

$$t_{f_2} \left(\frac{\lambda}{\mu} \right) = \begin{cases} \frac{\lambda(a+1)}{a\mu}, & \lambda \leq \frac{a^2\mu}{2(a+1)} \\ \sqrt{\frac{2\lambda(a+1)}{\mu}} - \frac{a}{2}, & \lambda > \frac{a^2\mu}{2(a+1)}. \end{cases}$$

Then, by (37), the optimal weighting parameter λ^* in each iteration of algorithm can be set as

$$\lambda_{k+1}^* = \begin{cases} \lambda^1 = \frac{av_{r+1}^{k+1}\mu_k}{a+1}, & \lambda^1 \leq \frac{a^2\mu_k}{2(a+1)} \\ \lambda^2 = \frac{(a+2v_{r+1}^{k+1})^2\mu_k}{8(a+1)}, & \lambda^1 > \frac{a^2\mu_k}{2(a+1)}. \end{cases} \quad (42)$$

- For $f_2(x) = \frac{a|x|}{1+a|x|}$, we have

$$t_{f_2} \left(\frac{\lambda}{\mu} \right) = \begin{cases} \frac{\lambda a}{\mu}, & \lambda \leq \frac{\mu}{2a^2} \\ \sqrt{\frac{2\lambda}{\mu}} - \frac{1}{2a}, & \lambda > \frac{\mu}{2a^2}. \end{cases}$$

Then, by (37), the optimal weighting parameter λ^* in each iteration of algorithm can be set as

$$\lambda_{k+1}^* = \begin{cases} \lambda^1 = \frac{v_{r+1}^{k+1}\mu_k}{a}, & \lambda^1 \leq \frac{\mu_k}{2a^2} \\ \lambda^2 = \frac{(2av_{r+1}^{k+1}+1)^2\mu_k}{8a^2}, & \lambda^1 > \frac{\mu_k}{2a^2}. \end{cases} \quad (43)$$

4.3 Adaptive algorithm

Using parameter-setting rules (31) and (38), the AIF algorithm will adapt to the choice of parameters μ and λ in each iteration. We call the AIF algorithm with parameter-setting rules (31) and (38) the adaptive alternating iterative framework (AAIF) algorithm which is described in Algorithm 2.

Algorithm 2 AAIF algorithm.

```

1: Input:  $\rho > 1, \mu_0 > 0$ 
2: Initialize:  $\mathbf{S}^0 \in \mathbb{R}^{m \times n}$ 
3: while not converged do
4:    $\mathbf{L}^{k+1} = \mathcal{H}_{f_1, \frac{1}{\mu_k}}(\mathbf{D} - \mathbf{S}^k)$ 
5:    $\lambda_{k+1}^* = \mu_k t_{f_2}^{-1}(v_{r+1}^{k+1})$  defined in (38)
6:    $\mathbf{S}^{k+1} = \mathcal{G}_{f_2, \frac{\lambda_{k+1}^*}{\mu_k}}(\mathbf{D} - \mathbf{L}^{k+1})$ 
7:    $\mu_{k+1} = \rho \mu_k$ 
8:    $k = k + 1$ 
9: end while
10: Output:  $\mathbf{L}^*, \mathbf{S}^*$ 

```

5 Numerical simulations

In this section, some numerical simulations are presented to validate the efficiency of AAIF algorithm in low-rank and sparse matrix decomposition. In AAIF algorithm, three different functions f_i are considered, that is, $f_1(x) = f_2(x) = |x|^{\frac{1}{2}}, \frac{(a+1)|x|}{a+|x|}, \frac{a|x|}{1+a|x|}$. For simplicity, the AAIF algorithm with $f_1(x) = f_2(x) = |x|^{\frac{1}{2}}$, $f_1(x) = f_2(x) = \frac{(a+1)|x|}{a+|x|}$ and $f_1(x) = f_2(x) = \frac{a|x|}{1+a|x|}$ are called AAIF-H algorithm, AAIF-T algorithm and AAIF-F algorithm, respectively. In numerical simulations, we will test the performance of these three algorithms and compare them with the classical IALM algorithm[8].

It is worth mentioning that the setting of the initial value μ_0 also has a significant impact on the performance of algorithms AAIF-H, AAIF-T and AAIF-F. In this paper, the initial parameter μ_0 is set as follows.

- In AAIF-H, the initial parameter μ_0 is set as

$$\mu_0 = \frac{\sqrt{54}}{4} (0.99 \|\mathbf{D}\|_2)^{-\frac{3}{2}}. \quad (44)$$

- In AAIF-T, the initial parameter μ_0 is set as

$$\mu_0 = \min \left\{ \frac{a+1}{0.99a \|\mathbf{D}\|_2}, \frac{2(a+1)}{(0.99 \|\mathbf{D}\|_2 + \frac{a}{2})^2} \right\}. \quad (45)$$

- In AAIF-F, the initial parameter μ_0 is set as

$$\mu_0 = \min \left\{ \frac{a}{0.99 \|\mathbf{D}\|_2}, \frac{2}{(0.99 \|\mathbf{D}\|_2 + \frac{1}{2a})^2} \right\}. \quad (46)$$

The algorithms are terminated by criterion

$$\text{RE} = \frac{\|\mathbf{D} - \mathbf{L}^k - \mathbf{S}^k\|_F}{\max\{\|\mathbf{D}\|_F, 1\}} \leq 10^{-7}$$

or maximum iteration steps 1000. The accuracy of the recovered low-rank matrix $\mathbf{L}^*$ is measured by

$$\text{RE}(\mathbf{L}^*) = \frac{\|\mathbf{L}^\sharp - \mathbf{L}^*\|_F}{\max\{\|\mathbf{L}^\sharp\|_F, 1\}},$$

and the accuracy of the recovered sparse matrix $\mathbf{S}^*$ is measured by

$$\text{RE}(\mathbf{S}^*) = \frac{\|\mathbf{S}^\sharp - \mathbf{S}^*\|_F}{\max\{\|\mathbf{S}^\sharp\|_F, 1\}}.$$

In simulations, the initial matrix $\mathbf{S}^0$ is set as the zero matrix. In AAIF-H, AAIF-T and AAIF-F, we set $\rho = 1.65$, and set μ_0 as (44), (45) and (46), respectively. As suggested in [15] and [16], the parameter a in AAIF-T is set 1, and in AAIF-F is set as 2.

5.1 Random low-rank and sparse matrix decomposition

In this subsection, we test the performance of the algorithms on some random low-rank and sparse matrices decomposition problems. We generate the observation matrix $\mathbf{D} \in \mathbb{R}^{m \times m}$ by $\mathbf{D} = \mathbf{L}^\sharp + \mathbf{S}^\sharp$, where $\mathbf{L}^\sharp$ and $\mathbf{S}^\sharp$ are the given low-rank matrix and sparse matrix. Given $r < m$, the low-rank matrix $\mathbf{L}^\sharp$ is generated by $\mathbf{L}^\sharp = \mathbf{N}_1 \mathbf{N}_2$, where $\mathbf{N}_1 \in \mathbb{R}^{m \times r}$ and $\mathbf{N}_2 \in \mathbb{R}^{r \times m}$ are the random matrices whose elements independently drawn by random from the norm distribution with zero mean and unit variance. The sparse matrix $\mathbf{S}^\sharp$ is constructed by setting a proportion of elements to be ± 500 and the rest to be zero. Given a proportion $sr \in [0, 1]$, we set the number of the nonzero elements of $\mathbf{S}^\sharp$ as $sr \times m^2$.

Tables 2 and 3 demonstrate the performance of algorithms on some random low-rank and sparse matrix decomposition problems with different sr ($sr = 0.20$ and $sr = 0.25$). From Tables 2 and 3, we can see that the algorithms AAIF-H, AAIF-T and AAIF-F perform effectively in decomposing the random low-rank and sparse matrices, and they all perform better than the convex optimization method IALM.

Table 2 Performance of algorithms in decomposing the random low-rank and sparse matrices, $m = 500$, $sr = 0.20$

Algorithm	r	$RE(L^*)$	$\text{rank}(L^*)$	$RE(S^*)$	$\ S^*\ _0$	k
AAIF-H	50	4.05e-06	50	9.95e-08	50000	36
	60	4.79e-06	60	1.35e-07	50000	38
	70	3.62e-06	70	1.13e-07	50000	42
AAIF-T	50	4.01e-06	50	1.01e-07	50000	36
	60	4.70e-06	60	1.33e-07	50000	39
	70	4.66e-06	70	1.47e-07	50000	43
AAIF-F	50	3.90e-06	50	9.70e-08	50000	36
	60	4.68e-06	60	1.32e-07	50000	39
	70	4.54e-06	70	1.43e-07	50000	43
IALM	50	2.64e-04	53	8.43e-06	52853	32
	60	2.70e-03	221	9.47e-05	114158	43
	70	5.53e-02	272	2.10e-03	142511	43

Table 3 Performance of algorithms in decomposing the random low-rank and sparse matrices, $m = 500$, $sr = 0.25$

Algorithm	r	$RE(L^*)$	$\text{rank}(L^*)$	$RE(S^*)$	$\ S^*\ _0$	k
AAIF-H	50	4.48e-06	50	1.04e-07	62500	40
	60	6.05e-06	60	1.58e-07	62500	43
	70	6.23e-06	75	1.86e-07	62500	48
AAIF-T	50	5.50e-06	50	1.29e-07	62500	41
	60	6.06e-06	60	1.58e-07	62500	45
	70	6.29e-06	92	1.86e-07	62500	50
AAIF-F	50	5.60e-06	50	1.31e-07	62500	41
	60	5.91e-06	60	1.56e-07	62500	45
	70	5.59e-06	74	1.65e-07	62500	51
IALM	50	3.48e-02	276	9.84e-04	141435	42
	60	1.22e-01	289	3.70e-03	147590	40
	70	1.99e-01	290	6.60e-03	149422	39

Fig. 1 Original Peppers image and its low-rank image. (a) original Peppers image, (b) rank 40 Peppers image

(a)



(b)

5.2 Image denoising

In the second numerical simulation, we test the performances of the algorithms for an image denoising problem. Consider a 256×256 grayscale image (Peppers), we first use the SVD to obtain its approximated low-rank image

with rank 40, and then use a sparse noise matrix to pollute the rank 40 Peppers image. The sparse noise matrix is constructed by random setting 10% of elements to be 100 and the rest to be zero. Original Peppers image, rank 40 Peppers image and the polluted image are displayed in Figs. 1 and 2, respectively.



Fig. 2 The rank 40 Peppers image polluted by sparse noise

Table 4 Performance of algorithms on image denoising

Algorithm	$RE(L^*)$	$\text{rank}(L^*)$	$RE(S^*)$	$\ S^*\ _0$	k
AAIF-H	9.72e-06	40	1.39e-07	6554	45
AAIF-T	3.64e-05	50	6.00e-07	6554	49
AAIF-F	2.20e-05	42	3.53e-07	6554	49
IALM	1.41e-02	122	2.33e-04	31962	129

From Table 4, we can see that all of AAIF-H, AAIF-T and AAIF-F perform better than IALM in decomposing the low-rank Peppers image and sparse noise. As we can see, the AAIF-H performs the best among these four algorithms in terms of $RE(L^*)$, $\text{rank}(L^*)$, $RE(S^*)$, $\|S^*\|_0$ and k , the AAIF-F performs the second best, and the AAIF-T performs the third best. The recovered low-rank Peppers images are displayed in Fig. 3.

5.3 Background modeling from surveillance video

To further test the performance of algorithms, we apply the algorithms to the surveillance video. In fact, surveillance video is a natural candidate for the low-rank and sparse matrix decomposition problem because of the correlation between frames. Given a sequence of surveillance video frames, we stack all video frames as columns of a matrix,

then the stationary background naturally corresponds to the low-rank component L_0 and the activities that stand out from the background naturally corresponds to the sparse component S_0 . In this subsection, the proposed algorithms are applied to a surveillance video to verify their effectiveness.

In this simulation, we consider a busy scene surveillance video which has a stationary background but the moving objects in the foreground. We use the first 200 frames and stack them as a matrix D . The size of each frame is 144×176 , and the size of the resulting matrix D is 25344×200 . In algorithms AAIF-H, AAIF-T and AAIF-F, the number of nonzero elements of sparse part (moving objects) is set as $0.0462 \times 2534 \times 200 \approx 23414$.

The convergence results of algorithms are presented in Table 5. In particular, we show in Fig. 4 the change of RE of algorithms AAIF-H, AAIF-T, AAIF-F and IALM with the iteration number increasing. From Fig. 4, we can see that the REs generated by algorithms converge to zero with the iteration number increasing. According to the results in Table 5 and Fig. 4, we can conclude that the proposed algorithms are all convergent.

From Table 5, we can further draw the following observations: (i) IALM has smaller iteration number and less computing time than AAIF-H, AAIF-T and AAIF-F; (ii) AAIF-H, AAIF-T and AAIF-F have the smaller RE than IALM, especially the AAIF-H has the smallest RE. By these two observations, IALM outperforms AAIF-H, AAIF-T and AAIF-F in iteration number and computing time. On the contrary, the AAIF-H, AAIF-T and AAIF-F outperform IALM in RE. Overall, AAIF-H, AAIF-T and AAIF-F perform better than the convex method IALM in RE, although they are all time-consuming than IALM. The decomposition results of the algorithms are displayed in Figs. 5, 6, 7 and 8, respectively.

6 Conclusion

In this paper, we study a generalized low-rank and sparse matrix decomposition problem for low-rank and sparse matrix decomposition by exploiting a family of nonconvex functions. The theoretical analysis of the generalized low-rank and sparse matrix decomposition problem has been established. It is proved that the generalized low-rank and sparse matrix decomposition problem can be exactly solved

Fig. 3 Low-rank Peppers images recovered by algorithms AAIF-H, AAIF-T, AAIF-F and IALM



(a) AAIF-H



(b) AAIF-T



(c) AAIF-F



(d) IALM

Table 5 Performance of algorithms on video decomposition

Algorithm	RE	k	Time(s)
AAIF-H	6.24e-08	43	20.76
AAIF-T	7.71e-08	38	18.23
AAIF-F	6.36e-08	37	16.67
IALM	8.04e-08	32	10.82

by solving the regularization generalized low-rank and sparse matrix decomposition problem. We present a tractable nonconvex algorithm framework to solve the regularization generalized low-rank and sparse matrix decomposition problem. A very powerful parameter-setting strategy is provided to adapt the optimal parameters in iteration of the proposed algorithm framework. Finally, some numerical simulations illustrate the performance of the proposed algorithms. The simulation results verified the validity of the proposed algorithms.

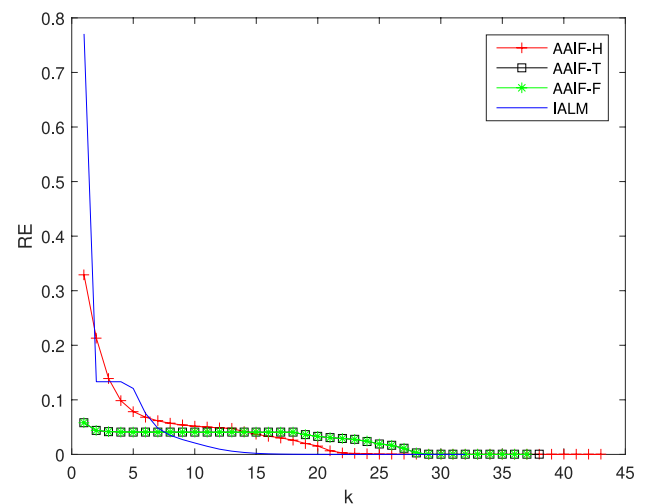


Fig. 4 The change of RE of algorithms with the iteration number increasing

Fig. 5 Decomposition results of AAIF-H. (a) The 100th frame surveillance video, (b) Background, (c) Moving objects



Fig. 6 Decomposition results of AAIF-T. (a) The 100th frame surveillance video, (b) Background, (c) Moving objects



Fig. 7 Decomposition results of AAIF-F. (a) The 100th frame surveillance video, (b) Background, (c) Moving objects



Fig. 8 Decomposition results of IALM. (a) The 100th frame surveillance video, (b) Background, (c) Moving objects



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Data Availability Data will be made available on reasonable request.

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